

Production Line Pressure Monitoring

Purpose

This work instruction defines the procedure for monitoring and recording pressure parameters on production lines to ensure process control and product quality.

Scope

This instruction applies to all production lines at Acme Corporation that use pneumatic or hydraulic pressure in their manufacturing processes.

Equipment Required

- Calibrated pressure gauge (range: 0-10 bar, accuracy: ± 0.1 bar)
- Digital pressure recorder or data logger
- Personal protective equipment: safety glasses, steel-toe boots

Procedure

Step 1: Pre-Shift Check

Before starting production each shift:

1. Verify the pressure gauge has a valid calibration sticker (calibrated within last 12 months).
2. Check that the gauge is properly connected and shows zero with no system pressure.
3. Record the gauge serial number, calibration date, and zero reading in the shift log.
4. If the gauge is out of calibration, notify the QA Supervisor immediately and do not use the line.

Step 2: Normal Operation Monitoring

During production:

1. Monitor the pressure gauge continuously during operation.
2. The acceptable pressure range is specified in the product-specific setup sheet (typical range: 4.5-5.5 bar).
3. Record pressure readings every 30 minutes in the pressure monitoring log.
4. Record the following for each reading:
 - Time
 - Pressure value (bar)
 - Initials of the operator
 - Any observations about process behavior

Step 3: Out of Specification Response

If pressure exceeds the upper specification limit or falls below the lower specification limit:

1. Immediately press the emergency stop button.
2. Record the out-of-spec pressure value and time.
3. Quarantine all products produced since the last acceptable reading.
4. Notify the QA Supervisor immediately (within 15 minutes).
5. Complete an NCR per SOP-QA-012.
6. Do not restart production until QA Supervisor authorization.

Step 4: Shift Handover

At the end of each shift:

1. Review the pressure log with the incoming operator.
2. Highlight any abnormal readings or trends.
3. Both operators sign the shift handover log.
4. If pressure showed an upward or downward trend, flag for QA review.

Acceptance Criteria

Parameter	Lower Limit	Upper Limit	Unit
Operating Pressure	4.0	5.5	bar
Pressure Variation (within shift)	—	± 0.3	bar
Pressure Drop Rate (idle)	—	0.1	bar/min

Related Documents

- SOP-QA-012: Non-Conformance Report Procedure
- WI-019: Calibration of Measurement Equipment
- Product Setup Sheets (per product code)

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